

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIC501 - DESIGN OF STEEL STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The structural advantage of using steel as a structural material is
 - a) Small weight-to-strength
 - b) Speed of erection
 - c) Speed of dismantling
 - d) Scrap value
- 2 As per the Indian Standard Institution, the types of rolled steel sections are
 - a) two
 - b) four
 - c) ten
 - d) twelve
- 3 Which of the following is correct for pitch of the bolts and gauge?
 - a) pitch is measured along direction of load, gauge is measured perpendicular to direction of load
 - b) pitch is measured perpendicular direction of load, gauge is measured along to direction of load
 - c) pitch is measured along direction of load, gauge is measured along to direction of load
 - d) pitch is measured perpendicular direction of load, gauge is measured perpendicular to direction of load

- 4 The maximum size of fillet weld is obtained by _____
- a) adding 1.5mm to thickness of thinner member to be jointed
 - b) adding 3mm to thickness of thinner member to be jointed
 - c) subtracting 3mm from thickness of thinner member to be jointed
 - d) subtracting 1.5mm from thickness of thinner member to be jointed
- 5 Which of the following is true statement?
- a) thickness of angle has no significant influence on member strength
 - b) net section efficiency is lower when long leg of angle is connected rather than short leg
 - c) when length of connection decreases, the tensile strength increases
 - d) effect of gusset plate thickness on ultimate tensile strength is significant
- 6 What is the effective length when both ends of compression member are fixed?
- a) $0.65L$
 - b) $0.8L$
 - c) L
 - d) $2L$
- 7 As per IS specification, the beam sections should be
- a) not symmetrical about any principal axes
 - b) at least symmetrical about one of the principal axes
 - c) symmetrical about all principal axes
 - d) unsymmetrical about all principal axes
- 8 For gantry girders carrying electrically operated overhead travelling cranes, the lateral forces are increased by ____ for impact allowance.
- a) 10% of weight of crab and weight lifted on the crane
 - b) 20% of weight of crab and weight lifted on the crane
 - c) 25% of maximum static wheel load
 - d) 50% of maximum static wheel load

- 9 What is the maximum vertical deflection allowed for a gantry girder where the cranes are manually operated?
 - a) $L/500$
 - b) $L/700$
 - c) $L/600$
 - d) $L/800$
- 10 The economic range of spacing of roof trusses is
 - a) $1/2$ to $1/3$ of span
 - b) $1/2$ to $1/4$ of span
 - c) $1/4$ to $1/6$ of span
 - d) $1/4$ to $1/5$ of span
- 11 The shape factor of an isosceles triangle for bending about the axis parallel to the base is
 - a) 1.5
 - b) 1.7
 - c) 2.0
 - d) 2.34
- 12 Plastic modulus for a circular section of diameter 'd' is
 - a) $d^3 / 6$
 - b) $d^3 / 3$
 - c) $d^3 / 4$
 - d) $d^3 / 5$

Q2. Solve the following.

- A) Explain Advantages and Disadvantages of steel structures. 6
- B) A 20 mm thick plate is joined to 18 mm plate by 200 mm long (effective) butt 6 weld. Determine the strength of joint if a Single V butt weld is used.

Q3. Solve the following.

- A) Design a single angle strut connected to the gusset plate to carry 180 KN axial compressive load. The length of the strut between center to center connections is 3m.
- B) Design a Simply Supported Beam of span 6m carrying a udl of 20 KN/m including self-weight. If the compression flange of the beam is laterally supported (or) restrained. Check the beam for moment carrying shear and deflection. The grade of steel is Fe 250.

Q4. Solve Any Two of the following.

A) Design angle purlin for the following data by simplified method:

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Spacing of trusses = 4 m

Spacing of purlins = 1.6 m

Weight of A.C sheets including laps and fixtures = 0.205 kN/m²

Live load = 0.6 kN/m²

Wind load = 1 kN/m²

Inclination of main rafter of truss = 21°

B) Calculate the maximum wheel load on gantry girder for the following data:

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a. Crane capacity = 100 kN

b. Self weight of crane girder excluding trolley = 150 kN

c. Self weight of trolley electric motor, hook etc = 35 kN

d. Approximate minimum approach of the crane = 1.0 m

e. Wheel base = 2.5 m

f. C/C distance between gantry rails = 20 m

g. C/C distance between column (span of gantry girder) = 7.5 m

h. Self weight of rail section = 300 N/m

i. Yield stress of steel = 250 N/mm²

C) List out the various elements of the roof truss and mark all its significance.

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Q5. Solve Any Two of the following.

A) Design a slab base for a column ISHB 300 @ 577 N/m carrying an axial compressive load of 1000 kN. M20 Concrete is used for the foundation. Provide welded connection between column and base plate.

B) Design a single lacing system for a column of effective length 9 m to support an axial compressive load of 1500 kN. Design the column which shall consist of two channels placed back to back at suitable spacing.

C) Explain in detail with neat sketches

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I) Slab Base

II) Gusseted Base

III) Moment Resisting Base

Q6. Solve Any Two of the following.

- A) Derive the relation between load factor and safety factor. 6
- B) A simply supported beam of span L is subjected to central point load W . 6

Find

- a. Shape factor of circular section
- b. Shape factor of Rectangular section
- C) Define Plastic Hinge. Explain its role in determining the collapse load of a steel structure. Support your answer with neat sketches.

*** End ***